

Total No. of Questions : 4]

SEAT No. :

PF283

[Total No. of Pages : 2

APR-26/TE/Insem-362

T.E. (Civil Engineering) (Insem)

DESIGN OF REINFORCED CONCRETE STRUCTURES

(2019 Pattern) (Semester - VI) (301013)

Time : 1 Hour 15 minutes]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1 or Q.2 and Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) IS 456 is permitted in the examination.
- 5) Non-programmable calculator is allowed in the examination.

Q1) a) A T-beam has the following details:

flange width = 1000 mm, web width = 250 mm, flange thickness = 100 mm, effective depth = 500 mm, area of reinforcement = 2200 mm². Use M20 concrete and Fe 500 steel. Determine moment of resistance assuming the flange in compression. **[8]**

b) Define the following terms used in Limit State Method: **[7]**

- i) Characteristic strength
- ii) Characteristic load
- iii) Partial safety factors for materials
- iv) Partial safety factors for loads.

Also explain why partial safety factors for loads are different for dead load, live load and wind/earthquake load.

OR

Q2) a) A doubly reinforced L-beam has the following details:

$b_f = 800$ mm, $b_w = 250$ mm, $t_f = 100$ mm, $d = 600$ mm, $d' = 50$ mm, $A_{st} = 2000$ mm² and $A_{sc} = 400$ mm². Calculate the moment of resistance using M30 grade concrete and Fe 500 grade steel. **[8]**

P.T.O.

b) Explain the following limit states and their importance in design: [7]

- i) limit state of collapse
- ii) limit state of serviceability
- iii) limit state of durability.

Q3) Design the slabs S-1, S-2, S-3 and S-4 shown in Fig. 1. Consider the width of the beam as 300 mm. The imposed load on the slabs S-1 to S-6 is 4 kN/m^2 and 2 kN/m^2 on S-7. Use M30 grade of concrete and Fe 500 grade steel.

The design should include all assumptions, design steps, checks for deflection and strength and details of reinforcement in the slabs. [15]

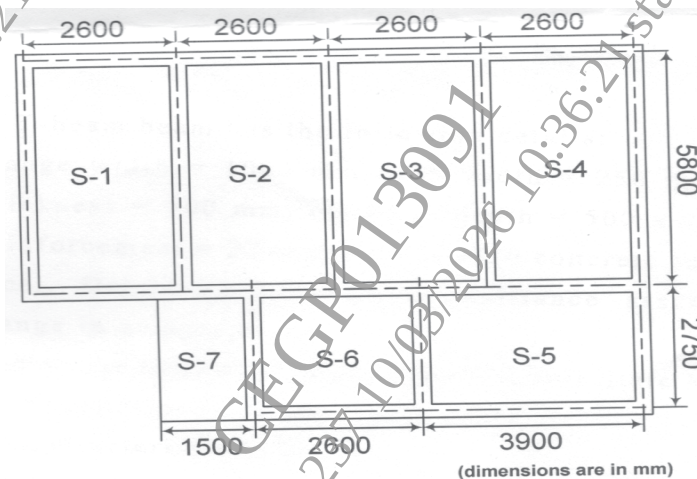


Fig. 1

OR

Q4) For the data given in Q.3, design the slabs S-7 and S-6. The design should include all assumptions, design steps, checks for deflection and strength and details of reinforcement in the slabs. [15]

